```
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PS C:\Users\DELL> nbtstat -a

Displays protocol statistics and current TCP/IP connections using NBT
(NetBIOS over TCP/IP).

NBTSTAT [ [-a RemoteName] [-A IP address] [-c] [-n]
        [-r] [-R] [-RR] [-s] [-S] [interval] ]

  -a   (adapter status) Lists the remote machine's name table given its name
  -A   (Adapter status) Lists the remote machine's name table given its
                        IP address.
  -c   (cache)          Lists NBT's cache of remote [machine] names and their IP addresses
  -n   (names)          Lists local NetBIOS names.
  -r   (resolved)       Lists names resolved by broadcast and via WINS
  -R   (Reload)         Purges and reloads the remote cache name table
  -S   (Sessions)       Lists sessions table with the destination IP addresses
  -s   (sessions)       Lists sessions table converting destination IP
                        addresses to computer NETBIOS names.
  -RR  (ReleaseRefresh) Sends Name Release packets to WINS and then, starts Refresh
```

```
File  Edit  View  Search  Terminal  Help
tudent@fedora: $ hostname
fedora
tudent@fedora: $
```

MATE Terminal

```
File  Edit  View  Search  Terminal  Help
tudent@fedora: $ arp -a
_gateway (10.0.2.2) at 52:55:0a:00:02:02 [ether] on enp0s3
tudent@fedora: $
```

```
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PS C:\Users\DELL> ipconfig /all

Windows IP Configuration

   Host Name . . . . . . . . . . . . : HDC0124096
   Primary Dns Suffix  . . . . . . . :
   Node Type . . . . . . . . . . . . : Hybrid
   IP Routing Enabled. . . . . . . . : No
   WINS Proxy Enabled. . . . . . . . : No
```

```
tudent@fedora: $ netstat
Active Internet connections (w/o servers)
Proto Recv-Q Send-Q Local Address           Foreign Address         State
udp        0      0 fedora:bootpc          _gateway:bootps         ESTABLISHED
Active UNIX domain sockets (w/o servers)
Proto RefCnt Flags       Type       State         I-Node   Path
unix  3      [ ]         STREAM     CONNECTED     31317    @/tmp/.X11-unix/X0
unix  3      [ ]         STREAM     CONNECTED     23637
unix  3      [ ]         STREAM     CONNECTED     20420    /run/user/1000/bus
unix  3      [ ]         STREAM     CONNECTED     18378    /run/dbus/system_bus_
socket
unix  3      [ ]         STREAM     CONNECTED     22517    /run/user/1000/bus
unix  3      [ ]         STREAM     CONNECTED     22255
unix  3      [ ]         STREAM     CONNECTED     24084
unix  3      [ ]         STREAM     CONNECTED     22770    /run/dbus/system_bus_
socket
unix  3      [ ]         STREAM     CONNECTED     23581    @/tmp/.X11-unix/X0
unix  3      [ ]         STREAM     CONNECTED     18504    /run/systemd/io.syste
n.ManagedOOM
unix  3      [ ]         STREAM     CONNECTED     24260
```

DATE:-
16.7.26

BASIC NETWORK COMMANDS

AIM :-

To execute the basic Network commands in windows for network configuration and trouble Shooting.

1. arp - a :-

ARP is short form of address resolution. protocol, It will show the IP computer along with the IP address and MAC address of your router.

2. Hostname :-

This is the simplest of all TCP/IP commands. It Simply displays the computer.

3. ipconfig /all :- This command displays detailed configuration information about connection including Router, gateway, DNS, DHCP, and type of Ethernet and System.

4. Nbstat - a :

This command helps solve problems with NetBIOS name resolution for netBIOS over TIC/IP

5. netstat - netStat displays a Variety of statistics about a TCP/IP connections. It is a command line tool for monitoring network connection incoming and outgoing as well as viewing routing tables,

```
Windows PowerShell
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PS C:\Users\DELL> nslookup
Default Server:  UnKnown
Address:  172.16.72.1
```

```
Windows PowerShell
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PS C:\Users\DELL> pathping

Usage: pathping [-g host-list] [-h maximum_hops] [-i address] [-n]
                [-p period] [-q num_queries] [-w timeout]
                [-4] [-6] target_name

Options:
    -g host-list      Loose source route along host-list.
    -h maximum_hops   Maximum number of hops to search for target.
    -i address        Use the specified source address.
    -n                Do not resolve addresses to hostnames.
    -p period         Wait period milliseconds between pings.
    -q num_queries    Number of queries per hop.
    -w timeout        Wait timeout milliseconds for each reply.
    -4                Force using IPv4.
    -6                Force using IPv6.
```

```
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PS C:\Users\DELL> Route

Manipulates network routing tables.

ROUTE [-f] [-p] [-4|-6] command [destination]
                  [MASK netmask]  [gateway] [METRIC metric]  [IF interface]

  -f            Clears the routing tables of all gateway entries.  If this is
                used in conjunction with one of the commands, the tables are
                cleared prior to running the command.

  -p            When used with the ADD command, makes a route persistent across
                boots of the system. By default, routes are not preserved
                when the system is restarted. Ignored for all other commands,
                which always affect the appropriate persistent routes.

  -4            Force using IPv4.

  -6            Force using IPv6.
```

```
Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\DELL> ping

Usage: ping [-t] [-a] [-n count] [-l size] [-f] [-i TTL] [-v TOS]
            [-r count] [-s count] [[-j host-list] | [-k host-list]]
            [-w timeout] [-R] [-S srcaddr] [-c compartment] [-p]
            [-4] [-6] target_name
```

interface statistics - eg: net stat -r

6. nslookup :- is a tool used to perform DNS lookups display DNS detail such as the IP address of a particular computer, the MX domain or the NS servers.

7. pathping :- pathping is unique to window's and is basically a combination of Tracert commands. pathping traces the route to the destination address then second test of each router along the way.

8. ping :-
command is the best way to test connective nodes.
ping use ICMP to communicate
1. # ping hostname (ping local host)
2. # ping ip address (ping 4.2.2.2)
3. # ping fully qualified domain name

9) Route :- route command is used to show the IP routing table. It is to setup static route to specific host or network via an interface.

RESULT :-

Thus the basic computer network commands are executed successfully using linux and windows.